

MOHAMED MUFASSIR

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PROFESSIONAL SUMMARY

Computer Systems Engineering undergraduate at SLIIT (2027) with hands-on experience in Digital Design, Embedded Systems, and Computer Architecture. Skilled in designing hardware-software systems spanning finite state machines, sequential logic circuits, PID control systems, and IoT-connected embedded applications. Experienced with ESP32, Raspberry Pi, and AVR microcontrollers across real-world projects. Seeking opportunities in FPGA, Digital Design, Embedded Systems, and Hardware Engineering.

EDUCATION

BSc (Hons) in Computer Systems Engineering

Nov 2023 – Expected Nov 2027

Sri Lanka Institute of Information Technology (SLIIT) | GPA: ~3.0

Relevant Modules: Digital Logic Design, Digital Systems Design, Real-Time Systems, Control Theory, Computer Architecture, Data Structures & Algorithms

TECHNICAL SKILLS

Digital Design & HW	Digital Logic Design, Finite State Machines (FSM), Sequential & Combinational Logic, Shift Registers, Cache Architecture, UART / SPI / I2C, Oscilloscope & Lab Instruments
Embedded Systems	ESP32, Raspberry Pi, AVR (ATmega), Embedded C, Sensors & Actuators, PCB Design
Programming	C/C++, Embedded C, Python, TypeScript, JavaScript, Java, Rust (Basic)
Software & Tools	Git, Linux, MATLAB, Proteus, EasyEDA, Fusion 360, VS Code

EXPERIENCE

Software Engineering Intern — ZILLIT (Startup)

Nov 2025 – Feb 2026

Nittambuwa, Sri Lanka

- Led sole development of a cross-platform POS desktop application using Electron.js and TypeScript, designed as part of a broader IoT ecosystem (desktop + mobile integration).
- Built the front-end of an enterprise financial management system using Vue.js, integrating with a Java Spring Boot backend.
- Collaborated on system architecture decisions for scalable, production-ready software in a fast-paced startup environment.

Freelance Web Developer — Self-Employed

2024 – Present

Local & Remote

- Designed and developed business websites for local clients including a jewelry shop, restaurant, and an indoor playground booking platform.
- Delivered end-to-end solutions covering UI design, branding, and front-end development using React, HTML/CSS, and JavaScript.

PROJECTS

Digital Logic Design Projects

2026

- Designed sequential logic circuits including a pseudo-random bit generator using XOR feedback and PISO shift register serialization.
- Implemented FSM-based sequence detectors with synchronized clocking, state transition analysis, and digital circuit verification.

Cache Memory Simulator

2025

- Developed a Python-based cache simulator to evaluate memory hierarchy performance, implementing configurable cache parameters.
- Analyzed cache hit/miss ratios across different access patterns, applying concepts from computer architecture and memory management.

PID-Based DC Motor Speed Control (Raspberry Pi + Python)

2026

- Designed a real-time closed-loop PID control system with encoder feedback for precise RPM measurement and regulation.
- Built a Python desktop GUI to visualize live motor behavior, RPM graphs, and PID tuning parameters in real time.

AVR Autonomous Line-Following Robot (ATmega / Arduino Uno)

2026

- Programmed an autonomous robot in Embedded C using FSM-based behavioral control with 3 IR sensors for line tracking and bay detection.
- Implemented dual-mode logic: crossline-triggered parking search with autonomous bay detection, and resume-on-double-crossline detection.

ESP32-CAM WiFi RC Car with Live Video Streaming

2026

- Built a WiFi-controlled RC car with live MJPEG streaming; ESP32-CAM served as both access point and video server.
- Developed a web-based controller (hosted on Vercel) enabling real-time directional control and camera feed from any networked device.

IoT Smart Surveillance & Access Control (ESP32)

2023

- Integrated camera module with Telegram Bot API for real-time intrusion detection and remote door lock/unlock commands.
- Implemented UART/I2C communication protocols for reliable real-time IoT messaging and remote control.

CERTIFICATIONS & LEARNING

Introduction to Embedded Systems Software and Development Environments — University of Colorado Boulder (Coursera)

Embedded software development workflows, GCC toolchain, Linux build environments, ARM-based embedded system concepts, version control with Git.

FPGA Design for Embedded Systems Specialization — University of Colorado Boulder (Coursera) – (Reading)

Understand and practice all aspects of FPGA development, including conception, design, implementation, and debugging.

INTEREST AREAS

Embedded Systems | Digital Design & FPGA Systems | Computer Architecture | IoT | Robotics & Control Systems | Industrial Electronics

REFERENCES

Prof. Anuradha Jayakody

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